

Unit 3

Motion in One and Two Dimensions

3.1 Uniformly Accelerated Motion in 1D

By the end of this section, you will be able to:

- Explain a uniformly accelerated motion in 1D;
- Explain the difference between average velocity and instantaneous velocity; and
- Solve problems involving average velocity, instantaneous velocity and acceleration.

3.2 Equations of Uniformly Accelerated Motion in 1D

By the end of this section, you will be able to:

- Derive and apply the equations of uniformly accelerated motion in 1D.
- Solve various problems of uniformly accelerated motion in 1d.

3.3 Graphical Representation of Uniformly Accelerated Motion in 1D

By the end of this section, you will be able to:

- Graphically, represent position versus time graph, displacement time graph, velocity time graph, and acceleration time graph of a uniformly accelerated motion in 1d.
- Interpret a graph of velocity vs. time, displacement vs. time and acceleration vs. time.
- Draw graphs from the kinematic equations.
- Determine the velocity of a motion from the slope of displacement-time graph, and acceleration from the slope of velocity – time graph.
- Determine the displacement covered from the area under velocity – time graph.

3.4 Vertical Motion

By the end of this section, you will be able to:

- Describe what vertical motion is.
- Solve various problems of vertical motion
- Describe the motion of objects that are in free fall.
- Calculate the time, position and velocity of an object given the initial velocity of an object under vertical motion
- Solve problems related to an object thrown vertically upwards, downwards, and a freely falling bodies.

3.5 Uniform Circular Motion

By the end of this section, you will be able to:

- Define a uniform circular motion
- Calculate for angular and tangential displacement, angular and tangential velocity, and centripetal acceleration,
- Explain uniform circular motion in the horizontal plane,
- Identify the force required for circular motion to take place